



Relationship between technology readiness, AI adoption and value creation in hospitality industry: Moderating role of technological turbulence

Syed Shah Alam^{a,*}, Husam Ahmad Kokash^a, Mst. Nilufar Ahsan^b, Saif Ahmed^c

^a Department of Marketing, College of Business Administration, Prince Sultan University, Riyadh 11586, Saudi Arabia

^b Faculty of Management, Multimedia University, Cyberjaya, Malaysia

^c Graduate School of Business, Universiti Kebangsaan Malaysia, UKM, Bangi, Selangor 43600, Malaysia

ARTICLE INFO

Keywords:

Artificial intelligence adoption
Value creation
Technology readiness
Dynamic capabilities
Technological turbulence

ABSTRACT

This study explores the relationship between technology readiness, AI adoption, dynamic capabilities, sustainable competitive advantage, and technological turbulence, assessing their collective impact on value creation within the Malaysian hospitality industry. Employing the Resource-Based View, Dynamic Capabilities Framework, and Contingency Theory, the research analyzes these strategic elements' roles in enhancing organizational performance. Using stratified purposive sampling, data was collected from 336 respondents. Structural equation modeling was utilized to analyze the relationships among the constructs. The results reveal that technology readiness and AI adoption positively influence value creation, with AI adoption playing a direct role in enhancing organizational capabilities and efficiencies. Interestingly, dynamic capabilities exhibited a complex impact, initially impeding value creation due to transitional challenges, but ultimately proving crucial in navigating technological turbulence. Sustainable competitive advantage emerged as a significant mediator, translating AI adoption and dynamic capabilities into enhanced value creation. However, the moderating effects of technological turbulence were selective, significantly influencing only the relationship between dynamic capabilities and value creation. These findings underscore the nuanced interplay between strategic resources and environmental factors in driving value creation in a technologically evolving industry.

1. Introduction

The hospitality industry stands at a pivotal juncture, melding traditional service values with the transformative potential of artificial intelligence (AI). As AI emerges as a crucial force, it drives unprecedented enhancements in service quality, operational efficiency, and personalized guest (Bulchand-Gidumal et al., 2023; Guo et al., 2024). However, the integration of AI demands a delicate balance of technology readiness and continuous adaptation to evolving market demands—areas that remain critically underexplored (Parvez, 2020; Warner and Wäger, 2019).

Recent developments underscore the urgency of addressing this balance. For example, Marriott International has been at the forefront of implementing AI-driven chatbots to handle customer inquiries, significantly enhancing response times and customer satisfaction (Al-Hyari et al., 2023). Similarly, Hilton has deployed AI-based systems for predictive maintenance in their properties (Takyar, 2024), resulting in a

notable reduction in operational disruptions and maintenance costs. These examples illustrate how AI is not only meeting the growing expectation for hyper-personalization but also establishing new benchmarks for service excellence in the industry. As reported by *The Wall Street Journal* and *CNBC* in 2023, such AI-driven innovations are increasingly becoming the standard in the competitive hospitality market (Captain, 2023; Siu, 2023).

Moreover, advancements in machine learning, a key subset of AI, have significantly improved data quality and algorithmic accuracy (Okafor et al., 2020; Whang et al., 2023). Despite these technological strides, the sector faces significant challenges in applying AI to enhance customer interactions and streamline operations (Bassano et al., 2020). For instance, the ethical implications of AI deployment, particularly regarding data privacy, have become a focal point of concern. Reports from *The Deloitte* reveal growing unease over AI's role in guest monitoring and data usage, sparking debates on how hospitality businesses can responsibly integrate these technologies without compromising

* Corresponding author.

E-mail addresses: salam@psu.edu.sa (S.S. Alam), hkokash@psu.edu.sa (H.A. Kokash), nilufar970@gmail.com (Mst.N. Ahsan), p120810@siswa.ukm.edu.my (S. Ahmed).

<https://doi.org/10.1016/j.ijhm.2025.104133>

Received 30 May 2024; Received in revised form 8 December 2024; Accepted 15 February 2025

Available online 25 February 2025

0278-4319/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

customer trust (Leila, 2024). For instance, the use of facial recognition technology in some hotels has raised questions about consent and data protection, prompting calls for clearer regulations and more transparent practices (Solarova et al., 2023).

Additionally, AI's potential to foster inclusive experiences and create new value—for instance, improving service accessibility for disabled guests—highlights its broader implications (Navarro et al., 2014; Vuong and Tran, 2021). AI-driven solutions, such as voice-activated room controls and tailored in-room services for guests with disabilities, are increasingly being adopted by hotels (Buhalis and Moldavska, 2021a). These innovations address a critical need for inclusivity in the industry. However, the unpredictable shifts characteristic of today's technological landscape complicates strategic AI deployment, blending preparedness with the necessity for agility (Arias-Pérez and Cepeda-Cardona, 2022). The competitive pressures resulting from technological disruption, as highlighted by *Harvard Business Review* and *TechCrunch* in 2024, further emphasize the need for businesses to swiftly adapt to maintain market relevance (Calder et al., 2024; Wiggers, 2024).

Research indicates that while hospitality businesses are poised to leverage their dynamic capabilities for AI adoption, there is a significant gap in harnessing these capabilities to fully engage stakeholders and enhance the guest experience comprehensively (Ghesh et al., 2023). The true challenge lies in utilizing AI not merely as a tool for operational efficiency (Rana et al., 2022) but as a strategic asset to actively generate value for users—a pathway crucial for achieving sustainable competitive advantage (Jiang and Wen, 2020; Kot and Leszczyński, 2022). This challenge is particularly pertinent as companies face mounting competitive pressures and the need for innovation to stay ahead of new market entrants rapidly embracing AI technologies.

To address these gaps, this study employs a quantitative approach using Partial Least Squares Structural Equation Modeling (PLS-SEM) to investigate how Malaysian hospitality businesses can leverage their dynamic capabilities for effective AI adoption. Specifically, the research seeks to answer the following questions:

1. How do technology readiness and dynamic capabilities influence AI adoption in Malaysia's hospitality industry, particularly in the context of technological turbulence?
2. How does AI adoption contribute to value co-creation between hospitality businesses and their customers, and how is this process influenced by the businesses' sustainable competitive advantage?
3. What is the mediating role of sustainable competitive advantage in the impact of technology readiness, dynamic capabilities, and AI adoption on value creation, and how does technological turbulence modify these relationships?

The anticipated outcomes of this study are set to challenge existing paradigms and extend the understanding of how dynamic capabilities are exploited in an AI-enhanced environment. The findings aim to provide hospitality businesses with a strategic blueprint for deploying AI, not merely to optimize operational efficiency but also to enrich the customer experience, thus redefining industry standards. Additionally, this research will inform policymakers and educators, guiding regulatory frameworks and academic curricula to prepare industry leaders for a future increasingly dominated by AI.

By examining these dynamics through a multifaceted theoretical lens, the research offers fresh insights into technology's role in fostering value creation and securing competitive advantage, significantly enriching the academic discourse on technology management in volatile environments. Moreover, exploring emerging AI technologies and their potential future roles in hospitality underscores the study's relevance in guiding long-term strategic planning and innovation.

2. Literature review

2.1. Integration of AI adoption, dynamic capabilities, and value co-creation in organizational performance

The existing literature offers extensive insights into AI adoption, dynamic capabilities, value co-creation, and sustainable competitive advantage, particularly within the hospitality, tourism, and SME sectors. The literature reveals several critical factors driving AI adoption across various industries. For instance, Islam et al. (2023) identify perceived ease of use, perceived usefulness, technology competency, top management support, and competitive pressure as significant influencers of AI adoption in Malaysia's tourism industry, with market turbulence playing a moderating role. Similarly, Lada et al. (2023) emphasize the importance of top management commitment and organizational readiness for AI adoption among Malaysian SMEs, noting that competitive pressure and employee adaptability have lesser impacts (see Table 1).

Dynamic capabilities have also been recognized as crucial for enhancing operational sustainability and firm performance, particularly in environments characterized by technological turbulence. Naz et al. (2023) demonstrate that big data analytics and AI capabilities significantly enhance hotel performance in Malaysia, moderated by environmental dynamism. Chatterjee et al. (2021) further explore how absorptive capacity and dynamic capabilities impact the adoption of ubiquitous customer relationship management (UCRM) systems, which, in turn, positively influence operational sustainability and firm performance.

AI's role in facilitating value co-creation is another significant theme in the literature. Carvalho and Alves (2022) provide a systematic review of customer value co-creation in the hospitality industry, identifying customer behaviors and technological resources as key drivers. Chandra and Rahman (2024) develop a framework integrating AI characteristics with customer behavior, showing how AI influences customer decision-making processes. Paschen et al. (2020) contribute to this discussion by examining the complex interactions between human and non-human actors in AI-enabled value co-creation.

Moreover, the impact of technological disruptions, including Industry 4.0 and the Metaverse, on the hospitality industry has been widely recognized. Buhalis et al. (2019), Buhalis et al. (2022), Buhalis and Moldavska (2021b) discuss the challenges and opportunities posed by these disruptions, emphasizing the need for strategic reengineering to maintain sustainable competitive advantage. Ed-Dafali et al. (2023) examine how Industry 4.0 readiness mediates the relationship between strategic orientations and sustainable competitive advantage for SMEs in emerging markets.

This study makes a significant contribution by integrating these diverse streams of literature to offer a comprehensive model that examines the interrelated impact of AI adoption, dynamic capabilities, and value co-creation on organizational performance and sustainability in Malaysia's hospitality and SME sectors. Unlike previous research that often treated these variables in isolation, this study investigates the interplay between AI adoption and dynamic capabilities, explores the moderating role of market turbulence and competitive pressure, and focuses on AI-facilitated value co-creation. By doing so, this research provides novel insights into how these elements collectively drive performance and strategic advantage in a rapidly evolving technological landscape, particularly within the unique context of Malaysia.

2.2. Theoretical underpinning

This study synthesizes insights from the Resource-Based View (RBV), Dynamic Capabilities Framework, and Contingency Theory to elucidate AI adoption and value creation in the hospitality industry, particularly within the context of technological turbulence. RBV emphasizes the strategic utilization of AI as a key resource that enhances sustainability and competitive advantage, with their effectiveness moderated by

Table 1
Literature Matrix.

Literature Source	Major Model Parameters	Findings
Islam et al. (2023)	Perceived ease of use, perceived usefulness, technology competency, top management support, competitive pressure, market turbulence	AI adoption in Malaysia's tourism industry is positively influenced by ease of use, usefulness, competency, management support, and competitive pressure. Market turbulence significantly moderates these relationships.
Lada et al. (2023)	Top management commitment, organizational readiness, competitive pressure, employee adaptability, external support	Top management commitment and organizational readiness are critical for AI adoption among Malaysian SMEs. Competitive pressure, employee adaptability, and external support have less significant impacts.
Naz et al. (2023)	Big data analytics capability, artificial intelligence capability, entrepreneurial orientations, environmental dynamism	AI and big data analytics capabilities enhance hotel performance in Malaysia, with environmental dynamism moderating the impact of big data analytics on co-innovation.
Ogbeibu et al. (2020)	Technological turbulence, green HRM, environmental dynamic capability, green team creativity, green product innovation	Technological turbulence and green HRM practices positively influence green creativity and innovation, while environmental dynamic capability negatively impacts green product innovation.
Senbeto and Hon (2021)	Technological turbulence, employee resilience, crisis leadership efficacy, employee openness, resistance to change	Technological turbulence positively impacts service innovation, mediated by employee openness and resistance to change. Crisis leadership efficacy strengthens these relationships.
Huang et al. (2021)	AI adoption susceptibility, AI technology types (e.g., search/booking engines, chatbots, virtual agents)	AI adoption susceptibility varies in the hospitality industry, with search/booking engines, chatbots, and virtual agents ranking high. A new evaluation framework is proposed for assessing AI adoption.
Chatterjee et al. (2021)	Absorptive capacity, dynamic capability, technology turbulence, UCRM adoption, operational sustainability, firm performance	Absorptive capacity and dynamic capability significantly impact UCRM adoption, which in turn positively affects operational sustainability and firm performance. Technology turbulence moderates these relationships.
Balasubramanian and Ragavan (2019)	Industry 4.0, global economic shifts, service innovation, technological disruption	The Malaysian hospitality industry faces challenges due to global economic shifts and technological disruption. The study provides innovative recommendations for stakeholders.
Pillai and Sivathanu (2020)	Perceived ease of use, perceived usefulness, perceived trust, perceived intelligence, anthropomorphism, technological anxiety	Perceived ease of use, usefulness, trust, and intelligence drive AI-powered chatbot adoption in India. Technological anxiety does not

Table 1 (continued)

Literature Source	Major Model Parameters	Findings
Kong et al. (2022)	AI research trends, technology acceptance, customer perception	significantly impact adoption intentions. AI research in hospitality has shifted focus from technology development to customer acceptance and perception, highlighting gaps in understanding AI's impact on customer experience.
Buhalis et al. (2019)	Technological disruptions, service innovation, smart environments	Rapid technological advancements are transforming tourism and hospitality, emphasizing the need for research into service innovations and the future impact of smart environments.
Lew et al. (2020)	Technology self-efficacy, perceived critical mass, mobile usefulness, mobile ease of use, mobile self-efficacy, perceived enjoyment	Mobile wallets' adoption in Malaysia's hospitality industry is driven by technology self-efficacy, usefulness, and enjoyment, explaining 61 % of the variance in adoption intentions.
Jabeen and Isakovic (2018)	Human knowledge, services, robotics applications, automation, AI implementation	Human knowledge and robotics applications are significant factors in AI and automation implementation in hospitality. A framework is proposed for sustainable strategy development.
Cain et al. (2019)	Robotics, AI, internal and external customer service, ethical issues	Robotics and AI are transforming customer service in hospitality, with significant implications for legal and ethical considerations. The study provides an overview of these transformations.
Goel et al. (2022)	Consumer adoption, AI and robotics, psychological, social, financial barriers	Various factors such as psychological and social barriers affect the adoption of AI and robotics in hospitality. The study offers an integrated antecedent-outcome framework.
Doborjeh et al. (2021)	AI methods, multimodal data sets, personalized AI models	AI methods are increasingly used in hospitality to predict tourism behavior, with a focus on multimodal data sets. The study proposes new personalized AI models for smart tourism platforms.
Solakis et al. (2022)	Value co-creation, customer perceptions, AI technologies (e.g., service robots, AI kiosks, chatbots)	Customer perceptions and AI technologies significantly influence value co-creation in tourism. The study provides a theoretical framework for future research on AI's role in value co-creation.
Chandra and Rahman (2024)	AI-facilitated value co-creation, customer decision-making, decision-making types	AI impacts value co-creation by influencing customer decision-making processes. The study develops a framework integrating AI characteristics with customer behavior.

(continued on next page)

Table 1 (continued)

Literature Source	Major Model Parameters	Findings
Paschen et al. (2020)	AI-enabled value co-creation, human and non-human actors, Service-Dominant Logic	AI enables value co-creation through complex interactions between human and non-human actors, highlighting the roles and resources exchanged in these interactions.
Carvalho and Alves (2022)	Customer value co-creation, customer behaviors, technological resources	Customer value co-creation is driven by behaviors, social environment, and technology, leading to enhanced organizational performance in hospitality.
Buhalis and Moldavska (2021b)	Voice assistants, customer service, operational capabilities, data security	Voice assistants in hospitality improve customer service and operational efficiency but raise significant data security and privacy concerns.
Buhalis et al. (2022)	Metaverse, hybrid experiences, hospitality ecosystem, strategic reengineering	The Metaverse presents opportunities for hybrid experiences in hospitality, requiring reengineering of business processes and strategies. The paper discusses the future impact of Metaverse on the industry.
Zhang et al. (2023)	Open innovation, organizational learning, knowledge management capability	Open innovation enhances organizational learning, contributing to sustainable competitive advantage. Knowledge management capability moderates this relationship.
Haseeb et al. (2019)	Social and technological challenges, sustainable competitive advantage, business performance	Social and technological challenges significantly impact sustainable competitive advantage and business performance for Malaysian SMEs. Strategic alignment is key.
Ed-Dafali et al. (2023)	Industry 4.0 readiness, strategic orientations, sustainable competitive advantage	Industry 4.0 readiness mediates the relationship between strategic orientations and sustainable competitive advantage for SMEs in emerging markets.
Acosta-Prado and Tafur-Mendoza (2022)	ICT, dynamic capabilities, sustainable performance	Dynamic capabilities mediate the relationship between ICT use/ acquisition and sustainable performance in technology-based firms.
Elgarhy and Abou-Shouk (2022)	Sustainable competitive advantage, entrepreneurial orientation, marketing capability, innovation capability	Sustainable competitive advantage partially mediates the impact of entrepreneurial and marketing capabilities on market performance in travel agencies.
Chaudhuri et al. (2022)	Digital technologies, economic and social value, entrepreneurial orientation	Adoption of digital technologies significantly impacts economic and social value creation for SMEs, with entrepreneurial orientation moderating these relationships.
Y. Sun et al. (2022)	AI innovation ecosystem, dynamic capabilities, innovation intelligibility	Value co-creation in the AI innovation ecosystem impacts competitive advantage and innovation

Table 1 (continued)

Literature Source	Major Model Parameters	Findings
Awamleh and Bustami (2022)	AI adoption, IT capabilities, competitive advantage, COVID-19 pandemic	Intelligibility through dynamic capabilities. IT capabilities mediate the relationship between AI adoption and competitive advantage, particularly during the COVID-19 pandemic.
Pratono (2021)	Entrepreneurial culture, product development, technological turbulence	Entrepreneurial culture is crucial for achieving sustainable competitive advantage through product development, especially under technological turbulence.
Mishra and Pani (2020)	Organizational Learning, Ambidexterity, Research and Development, Value Creation, Servicization, Open Innovation	The roadmap highlights the role of organizational learning and ambidexterity in balancing innovation and operational efficiency, emphasizing AI's potential for product and process transformation. Servitization and open innovation are identified as critical strategies for value creation and sustainable competitive advantage.

leadership and external market conditions (Chaudhuri et al., 2022; Cheah et al., 2018; Parvez, 2020). Central to this view is the assertion that AI, as a strategic technological resource, can significantly enhance performance and sustainability when properly managed amidst technological turbulence and with effective leadership support (Chatterjee et al., 2021; Zahra, 2021).

The Dynamic Capabilities Framework complements RBV by stressing the need for ongoing adaptation and innovation in response to rapid technological changes (Dhanu et al., 2024; Giannoukou, 2024). This framework highlights the necessity for businesses to adapt their models swiftly and effectively in response to environmental changes such as market fluctuations and crises, ensuring competitive advantage and continuous innovation (Campo et al., 2014; Cheah et al., 2018; Linde et al., 2021). Additionally, this framework underscores the industry's transition towards e-business models through strategic planning and learning orientations, which are crucial for leveraging AI in enhancing operational efficiency and customer engagement (Pernsteiner and Rauseo, 2000; Roy and Pagaldiviti, 2023).

Meanwhile, Contingency Theory posits that optimal strategies are contingent upon fluctuating internal and external circumstances, making it crucial for organizations to navigate disruptions and enhance service innovation in the hospitality sector (Buhalis and Sinarta, 2019; Gaur et al., 2021; Senbeto and Hon, 2021). This theory provides a lens through which the alignment of AI technologies with organizational strategies is examined, assessing how this synergy can propel service innovation and heighten competitive advantage (Abedin, 2022). The theory further supports the industry's adoption of smart environments and advanced technologies like AI and robotics, which are vital for addressing challenges such as health crises and ensuring touchless travel (Drexler and Beckman Lapré, 2019; Gaur et al., 2021). This integration is particularly critical in navigating the complexities of market conditions and organizational readiness, which are key for achieving sustainable success in a competitive industry.

Collectively, these theoretical perspectives provide a robust framework (refer to Fig. 1) for understanding how leveraging advanced technologies and strategic adaptability are essential for maintaining competitiveness and fostering value creation in a dynamic and rapidly evolving sector. By combining RBV, Dynamic Capabilities Framework,

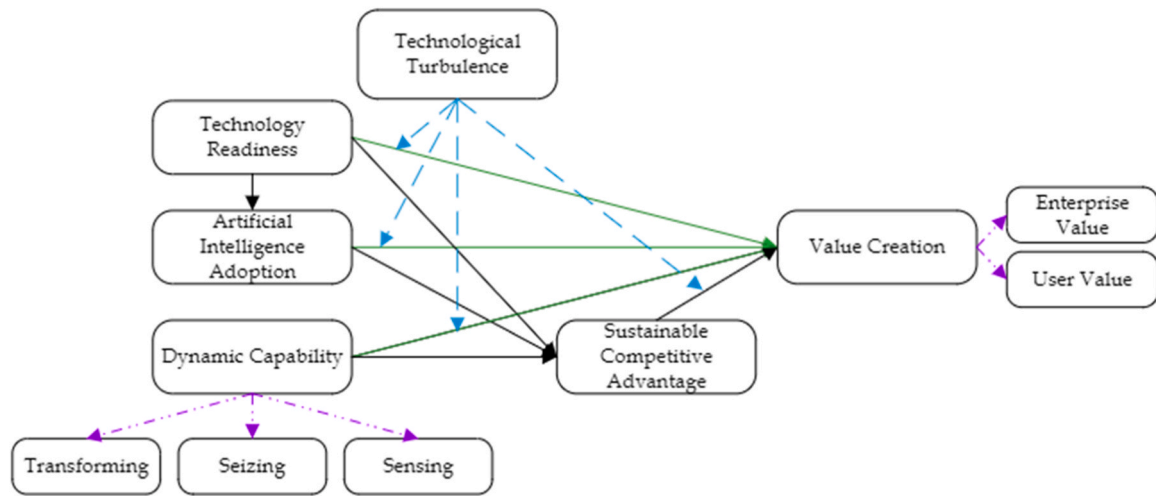


Fig. 1. Research Framework.

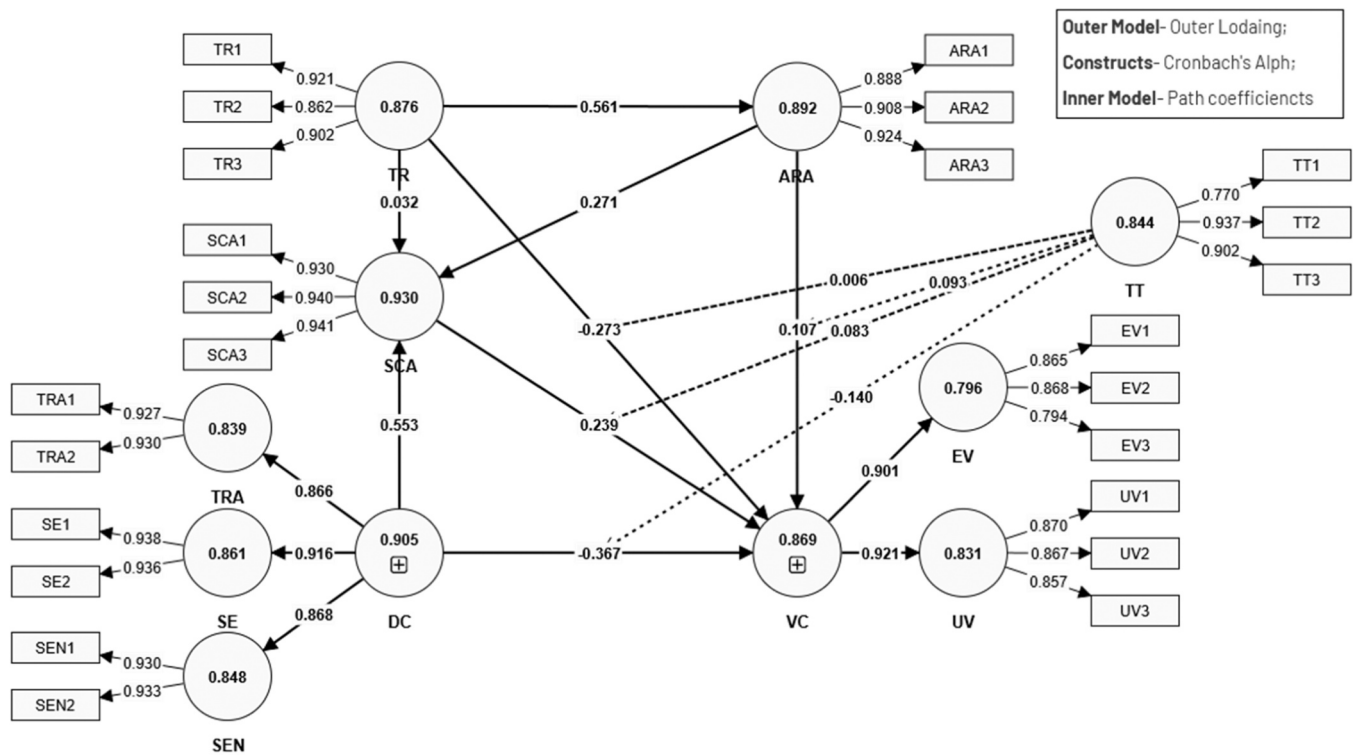


Fig. 2. Measurement Model.

and Contingency Theory, this study aims to offer a comprehensive understanding of how AI can be strategically managed to achieve long-term success in the hospitality industry.

2.3. Hypothesis development

Damerji and Salimi (2021) and Sudaryanto et al. (2023) established that perceived ease of use and usefulness significantly affect AI adoption, though Sudaryanto noted that technology readiness might not directly impact AI adoption. Further supporting this, Palade and Carutasu (2023) and AlSheibani et al. (2018) suggested that AI readiness fits well within broader digitization frameworks, while Flavián et al. (2021) and Godoe and Johansen (2012) underscored the crucial role of technology readiness, particularly through customers' technological

optimism and discomfort. Jöhnk et al. (2021) identified key AI readiness factors such as assets, capabilities, and commitment.

Regarding technology readiness and value creation, Kearney et al. (2018) and White et al. (2017) found that technology readiness enhances value creation behaviors for both enterprise and users, supported by Kearney's emphasis on digital technology satisfaction. Fujihara et al. (2022) and Lin (2007) noted that human values and technology readiness influence online purchase intentions and perceived e-service value, respectively. Similarly, Lee et al. (2009) linked personal values and technology readiness to service quality and customer loyalty, with Tuyet and Tuan (2020) and Yieh et al. (2012) confirming the impact on perceived value and customer satisfaction.

These findings lead to the formulation of the following hypotheses:

H1. . Technology readiness positively influences AI adoption within

organizations.

H2. . Technology readiness positively impacts value creation for both enterprises and users.

AI adoption significantly influences value creation across various sectors (Bassano et al., 2020; Haftor et al., 2024; Kot and Leszczyński, 2022; Payne et al., 2021; Washizaki, 2020; Y. Zhu et al., 2022). In the UAE, organizational decisions are shaped by the accessibility of AI technologies, with perceived risks associated with job displacement and creation being notable factors (Alzaabi and Shuhaiber, 2022). In healthcare, the value creation facilitated by AI hinges on digital self-efficacy and the quality of relational services, with shared decision-making as a pivotal component (Swan et al., 2023). Perifanis and Kitsios (2023) suggests that integrating AI into business and IT strategies offers potential for developing new business models and gaining competitive advantages. In the luxury sector, AI applications such as chatbots contribute to value creation by enhancing customer engagement (Bassano et al., 2020). Moreover, the appeal of intelligent personal assistants is driven by their usefulness, enjoyment, technical features, and cost benefits, which collectively influence their perceived value and adoption (White et al., 2017). AI virtual humans create value by fulfilling social and respectful needs through interactive experiences (Xu, 2023).

Dynamic capabilities also play a crucial role in value creation, as evidenced by multiple studies. Rashidirad and Salimian (2020a) and Dyduch et al. (2021) identified that sensing, learning, integrating, and coordinating capabilities are essential for SMEs to create value. Yu (2021) found that dynamic capabilities positively moderate the impact of enterprise digitalization on financial performance. X. Sun and Zhang (2021) and Madhala et al. (2020) highlighted the importance of resource and technology integration capabilities, along with analytics, in steering clear of value creation pitfalls and achieving data-driven outcomes for both enterprise and users. Innovation and effective decision-making capabilities are key to enhancing value creation and competitive advantage, as noted by Ayegba and Lin (2020) and Mehralian et al. (2022). Additionally, Sasmoko et al. (2019) emphasized the critical role of digital leadership in boosting innovation and dynamic capabilities.

These insights substantiate the following hypotheses:

H3. . AI adoption will positively influence value creation.

H4. . Dynamic capabilities will positively influence value creation.

Numerous studies have investigated the link between sustainable competitive advantage (SCA) and value creation. Gupta and Benson (2011) found that sustainable companies are not only competitive but also excel financially. Hariastuti et al. (2021) identified key drivers of sustainable value creation, such as environmental and social factors, and their impact on competitive advantages. Bhattacharjee (2018) explored how disruptive value creation affects competitive forces, while Rodriguez et al. (2002) proposed a dynamic firm perspective incorporating sustainable development. Laszlo et al. (2006) stressed the importance of stakeholder integration in decision-making to enhance competitive advantage. Empirical findings by Hahn et al. (2007) in the manufacturing sector and a demand-based view of SCA by Adner and Zemsky (2004) further validate the significant relationship between SCA and value creation, which is echoed by Thai and Mai (2023) who linked competitive advantage to sustainable growth.

H5. . Sustainable competitive advantage will mediate the positive relationship between technology readiness and value creation.

The mediation role of SCA in the relationship between technology readiness and value creation has been supported by various studies. Talebnejad (2012) emphasized the importance of managerial skills in IT for creating and sustaining SCA, which is crucial for value creation as suggested by Adner and Zemsky (2004). Gupta and Benson (2011) and

Rodriguez et al. (2002) highlighted that sustainable companies, often adept at leveraging technology, excel competitively and financially, with sustainability enhancing SCA and competitiveness. The digital economy further leverages technology for sustainability-driven competitive advantage (Nurova and Freze, 2021).

H6. . Sustainable Competitive Advantage mediates the influence of technology readiness on Value Creation.

The role of AI in value creation is intricate and influenced by multiple factors. Studies like Perifanis and Kitsios (2023) and Khan et al. (2024) indicate AI's significant impact on business value across areas such as decision-making and supply chain optimization. Yet, its influence is mediated by factors like AI availability and perceived risks (Alzaabi and Shuhaiber, 2022; Govindan, 2024). The interplay of AI and sustainability, enhancing green innovation and Sustainable Development Goals (Mhlanga, 2021; Wang, 2023), suggests SCA's mediating role, supported by strategic frameworks like those proposed by Bari et al. (2024) which advocate for SCA through sustainable dynamic capabilities.

H7. . Sustainable Competitive Advantage mediates the influence of AI adoption on Value Creation.

Furthermore, dynamic capabilities (DC) are pivotal in sustaining competitive performance and advantage, as highlighted by Mikalef and Pateli (2017) and Acosta-Prado and Tafur-Mendoza (2022), who underline the essential role of IT-enabled dynamic capabilities. Correia et al. (2020) suggests that dynamic capabilities mediate the relationship between market orientation and competitive advantages, affecting business performance and linking DC directly to value creation through SCA.

H8. . Sustainable Competitive Advantage mediates the influence of Dynamic Capability on Value Creation.

Low et al. (2021) and Chng et al. (2021) highlight the significant moderating role of technological turbulence, where Lee emphasizes its impact on internal market orientation processes, and Ch'ng demonstrates its positive influence on eco-organizational innovation and social performance. The dynamic nature of technological turbulence implies that firms operating in turbulent environments need to be more adaptive and flexible to maintain a competitive edge. This aligns with Cheah et al. (2018) and Matraves and Rondi (2005) findings that industry turbulence can enhance a firm's Sustainable Competitive Advantage (SCA) by forcing companies to continuously innovate and differentiate their products.

The role of digitalization and disruptive value creation in sustaining competitive advantages under technological turbulence is further supported by Knudsen et al. (2021) and Bhattacharjee (2018). They suggest that firms that successfully navigate digital transformation in turbulent environments can leverage these disruptions to create sustained competitive advantages. Furthermore, strategic responses such as maneuverability and flexibility, as discussed by Mariam et al. (2023) and Celuch and Kasouf (2014), illustrate how firms can innovate business models and enhance customer value despite—or because of—turbulent conditions.

Dayioglu and Kuskul (2022) and Wiesel (2022) provide additional insights into the strengthening effect of technological turbulence on aligning organizational commitment with value creation activities and on customer-based corporate valuation. Margarida et al. (2018) explore how technological turbulence influences the relationship between organizational innovation intensity and export performance, particularly through the mediating role of marketing capabilities. These insights underscore the critical role that technological turbulence plays in shaping firm strategies and outcomes.

Finally, studies by Autry et al. (2010) and Hamdi et al. (2016) highlight how technological turbulence moderates key organizational processes, such as technology acceptance and innovation speed, further

emphasizing its pivotal role in different organizational contexts.

Given this theoretical background, the following hypotheses are proposed:

H9. . *Technological Turbulence moderates the relationship between Sustainable Competitive Advantage and Value Creation.*

H10. . *Technological Turbulence moderates the relationship between technology readiness and Value Creation.*

H11. . *Technological Turbulence moderates the relationship between AI adoption and Value Creation.*

H12. . *Technological Turbulence moderates the relationship between Dynamic Capability and Value Creation.*

3. Research methodology

The choice of Malaysia as the focus for this study on AI adoption and value creation in the hospitality industry is motivated by its vibrant hospitality sector, burgeoning AI ecosystem, and strategic significance in the Asia-Pacific region. Malaysia's commitment to technological innovation, supported by a significant rise in AI startups, provides a dynamic backdrop for exploring AI's integration across diverse hospitality settings. This setting allows for a nuanced examination of how advanced technologies can enhance service delivery and operational efficiency.

3.1. Sampling

Utilizing a stratified purposive sampling approach, the research is designed to capture diverse perspectives across various industry management levels, focusing on prominent hospitality companies and AI startups in Malaysia. The data collection period of December 2023 to February 2024 was strategically chosen to align with the year-end and new-year business planning phases, when organizations typically make or evaluate decisions on technological adoption, including AI. This timing also coincided with a post-peak tourism season in Malaysia, ensuring better availability of key industry stakeholders.

Out of the initial group, 179 organizations showcased extensive AI exposure within their operations, a criterion crucial for ensuring the data accurately reflected deep-seated experiences and perspectives on AI's implementation and impact. Ultimately, 84 organizations, encompassing both startups and established entities, participated and returned completed questionnaires from four designated respondents each, totalling 336 individual responses. The respondent stratification within each organization included Top-Level Management, Middle Management, Operational Management, and Technical Specialists. Prior to the main survey, a pilot test involving 20 selected respondents with significant AI exposure was conducted to validate the questionnaire's reliability and clarity.

3.2. Measurement items development

Dynamic Capability is conceptualized based on the pioneering frameworks of Teece, Pisano (1994), and further refined by Eisenhardt and Martin (2000). This construct encompasses three core dimensions: a. Sensing- detecting and perceiving opportunities and threats, b. Seizing- mobilizing resources to address these opportunities and threats, and c. Transforming- the ongoing adaptation and realignment of the organization (Ma et al., 2021). Value creation is framed within the bifurcates into Enterprise Value (organizational benefits from collaborative endeavors) and User Value (the personalized satisfaction and value derived by the users) (Chuang and Lin, 2015). Technology Readiness, as outlined by Shehadeh et al. (2022), evaluates an organization's infrastructure, cultural orientation, and strategic inclination towards embracing emerging technologies like AI, analytics, and cloud computing, highlighting the proactive adoption of digital innovations

for business enhancements. Artificial Intelligence Adoption, guided by Chatterjee et al. (2023), probes into the extent and efficacy of AI technology utilization within hospitality operations, emphasizing AI's role in automating processes and augmenting decision-making with superior computational intelligence. Sustainable Competitive Advantage, drawing from Kim et al. (2011), investigates the long-term strategic assets that facilitate superior organizational performance. This construct focuses on leveraging AI and digital technologies for innovation, market development, and resolving industry challenges. Technological Turbulence, also based on Kim et al. (2011), captures the dynamic and often disruptive nature of technological evolution within the industry, examining its influence on strategic innovation and organizational responsiveness to technological breakthroughs.

3.3. Data analysis technique

In this study, the data analysis method utilized IBM SPSS for descriptive and demographic statistics to understand the sample's characteristics, and SmartPLS v 4.0.8.1 software for structural equation modeling (SEM), aligning with Hair et al. (2022) recommendations. This two-stage SEM approach began with establishing the reliability and validity of all constructs, including the intricate dimensions of Value creation and Dynamic Capability as higher-order constructs. It then rigorously examined the relationships among these constructs within the Malaysian hospitality industry's context of AI adoption, ensuring a thorough investigation of their multidimensional roles and impacts.

4. Data analysis

The demographic analysis from this study on AI adoption and value creation for both enterprise and users within the Malaysian hospitality industry reveals a predominantly young and well-educated workforce, with the majority aged between 25 and 44 years and holding at least a Bachelor's degree. This suggests a workforce that is likely adaptable and receptive to new technologies, including AI. The sample shows a higher proportion of male participants but also significant female representation, indicating diversity in the industry's approach to technology adoption. With an even distribution across various organizational roles—from top-level management to technical specialists—and a strong inclination towards AI startups (88 % of respondents), the data points to a dynamic sector keen on innovation. Additionally, the wide range of experience in the hospitality industry, predominantly less than 10 years, coupled with a substantial engagement in AI projects (86 % of organizations involved to some extent), underscores a forward-looking industry prepared for digital transformation. This blend of youthful enthusiasm, educational sophistication, and balanced insight across management levels presents a fertile ground for exploring the transformative potential of AI in enhancing operational efficiencies and creating value within the hospitality sector.

4.1. Measurement model statistics

In examining the measurement model, for robustness of which is confirmed through an analysis of measurement item statistics (Table 2) and an evaluation of discriminant validity (Tables 3 and 4). This comprehensive approach demonstrates the model's capability to capture the complexity of constructs such as Dynamic Capability and Value creation, essential for dissecting the interactions between technology readiness, artificial intelligence adoption, sustainable competitive advantage, and technological turbulence.

Measurement item statistics from Table 2 indicate strong internal consistency and reliability across all constructs. For example, Artificial Intelligence Adoption (ARA) showcases an Average Variance Extracted (AVE) of 0.822, and Sustainable Competitive Advantage (SCA) posts an AVE of 0.878, indicating significant convergent validity (Hair et al., 2022). Furthermore, the Variance Inflation Factor (VIF) values,

Table 2
Measurement Items Statistics.

Contact	Code	OL	VIF	CA	CR (rho_c)	AVE
ARA	ARA1	0.888	2.342	0.892	0.933	0.822
	ARA2	0.908	2.755			
	ARA3	0.924	3.083			
EV	EV1	0.865	1.843	0.930	0.956	0.878
	EV2	0.868	1.915			
	EV3	0.794	1.499			
SCA	SCA1	0.930	3.516	0.861	0.935	0.878
	SCA2	0.940	3.900			
	SCA3	0.941	3.953			
SE	SE1	0.938	2.336	0.848	0.930	0.868
	SE2	0.936	2.336			
SEN	SEN1	0.930	2.186	0.876	0.924	0.802
	SEN2	0.933	2.186			
TR	TR1	0.921	2.984	0.839	0.926	0.862
	TR2	0.862	1.973			
	TR3	0.902	2.758			
TRA	TRA1	0.927	2.097	0.844	0.905	0.762
	TRA2	0.930	2.097			
TT	TT1	0.770	1.596	0.831	0.899	0.748
	TT2	0.937	2.863			
	TT3	0.902	2.672			
UV	UV1	0.870	1.937	0.859	0.918	0.872
	UV2	0.867	1.945			
	UV3	0.857	1.856			

Table 3
Discriminate Validity (HTMT).

	EV	SCA	SE	SEN	ARA	TR	TRA
EV							
SCA	0.800						
SE	0.718	0.834					
SEN	0.554	0.710	0.839				
ARA	0.708	0.815	0.888	0.771			
TR	0.518	0.632	0.716	0.802	0.634		
TRA	0.733	0.796	0.820	0.709	0.838	0.64	
TT	0.837	0.843	0.848	0.801	0.829	0.831	0.844
UV	0.809	0.811	0.822	0.726	0.805	0.636	0.825
TT x TR	0.209	0.286	0.360	0.441	0.281	0.584	0.285
TT x ARA	0.213	0.378	0.471	0.458	0.433	0.346	0.371
TT x DC	0.256	0.389	0.496	0.52	0.422	0.433	0.378
TT x SCA	0.274	0.456	0.456	0.430	0.404	0.377	0.395
TT							
UV	0.847						
TT x TR	0.458	0.280					
TT x ARA	0.439	0.382	0.746				
TT x DC	0.485	0.405	0.844	0.823			
TT x SCA	0.483	0.440	0.765	0.855	0.818		

Table 4
Discriminant Validity (FLC).

	EV	SCA	SE	SEN	ARA	TR	TRA	TT	UV
EV	0.843								
SCA	0.691	0.937							
SE	0.597	0.773	0.937						
SEN	0.457	0.632	0.717	0.932					
ARA	0.598	0.743	0.777	0.670	0.907				
TR	0.432	0.572	0.623	0.692	0.561	0.895			
TRA	0.601	0.704	0.697	0.599	0.725	0.550	0.928		
TT	0.713	0.861	0.822	0.752	0.821	0.770	0.888	0.873	
UV	0.660	0.803	0.697	0.610	0.751	0.543	0.690	0.847	0.865

comfortably below the critical value of 5 (Thompson et al., 2017), confirm the absence of multicollinearity, ensuring the distinctiveness and integrity of each construct.

The study's discriminant validity, rigorously assessed through the HTMT criterion in Table 3 and the FLC criterion in Table 4, reinforces the model's credibility. HTMT values remain below the conservative threshold, and FLC scores confirm that the constructs share more variance with their indicators than with one another, ensuring the uniqueness of each construct's contribution to our understanding of the technological and strategic landscape within the industry (Shiau et al., 2019).

The analysis of higher order constructs within the study reveals strong and statistically significant loadings, as evidenced by the results presented in Table 5. The outer loading statistics demonstrate robust relationships between higher order constructs and their respective sub-dimensions. Specifically, Enterprise Value (EV) and User Value (UV) as dimensions of Value Creation (VC) show high loadings of 0.896 and 0.925 respectively, both with extremely significant p-values ($p < 0.001$), indicating a strong validation of the VC construct in the context of AI-driven enhancements in hospitality. Similarly, the sub-dimensions of Dynamic Capability (DC)—Sensing (SEN), Seizing (SE), and Transforming (TRA)—exhibit substantial loadings of 0.859, 0.918, and 0.872 respectively, further substantiating the effectiveness of dynamic capabilities in leveraging AI technologies for operational and strategic innovations in the sector. These statistical outcomes not only confirm the reliability and validity of the constructs within our model but also highlight the significant impact of AI adoption on enhancing both the operational and user-centric aspects of value creation in the hospitality industry.

By delving into dimensions such as Sensing, Seizing, and Transforming under Dynamic Capability, and distinguishing between Enterprise Value and User Value within Value creation, the study adeptly highlights how hospitality organizations in Malaysia navigate and leverage AI for competitive advantage and Value creation. In essence, the development and validation of this sophisticated measurement model, as detailed through Tables 1, 2, 3, and 4 solidify the study's capacity to explore the synergistic relationship between AI adoption and value creation in the hospitality sector. The constructs' statistical

Table 5
Outer Loading statistics of higher order constructs.

	Original sample	Sample mean	Standard deviation	T statistics	P values
EV < - VC	0.896	0.895	0.014	62.141	0.000
UV < - VC	0.925	0.925	0.007	134.126	0.000
SE < - DC	0.918	0.918	0.011	86.95	0.000
SEN < - DC	0.859	0.857	0.029	30.065	0.000
TRA < - DC	0.872	0.872	0.015	56.629	0.000

strengths, including high loadings and composite reliability scores across the board, illustrate a well-prepared industry for digital transformation. The findings not only contribute a refined analytical framework to the academic discourse but also offer actionable insights for industry stakeholders aiming to harness AI's transformative potential within the dynamic landscape of the Malaysian hospitality industry.

4.2. Structural model statistics

The structural model analysis, as illustrated in Fig. 3 and detailed through the Structural Model Statistics (Table 6), reveals insightful findings on the influence of higher-order constructs like Dynamic Capability (DC) and the interactions between technology readiness (TR), artificial intelligence adoption (ARA), sustainable competitive advantage (SCA), and technological turbulence (TT) on value creation (VC) in the Malaysian hospitality industry.

The data shows a strong and significant relationship between technology readiness (TR) and AI adoption (ARA), with a beta coefficient of 0.561 and a p-value below 0.001. This indicates that organizations with higher technological readiness are more likely to adopt AI technologies effectively, explaining about 31.5 % of the variance in AI adoption. However, the direct relationship between Technology Readiness (TR) and Value Creation (VC) is unexpectedly negative ($\beta = -0.275$, $p < 0.001$), indicating that technological readiness alone may not ensure value creation. This counterintuitive finding suggests that transitional challenges, such as misaligned implementation strategies or ineffective integration with existing processes, could impede the realization of value. A more detailed exploration of these transitional challenges and their implications is provided in the discussion section.

AI adoption itself positively influences value creation ($\beta = 0.110$, $p = 0.034$), though the effect size is modest, indicating that while AI contributes to value, its impact may depend on other factors. Conversely, dynamic capability (DC) shows a negative relationship with value creation ($\beta = -0.385$, $p < 0.001$), highlighting possible challenges in adapting capabilities effectively in turbulent environments. This suggest that external pressures, resource misallocation, or strategic missteps may counteract the expected benefits of dynamic capabilities. This issue is further elaborated upon in the discussion section to contextualize the challenges and provide theoretical and practical

Table 6
Structural Model Statistics.

Relationship Path	Original sample	T statistics	P values	F ²	R ²	R ² adjusted
TR -> ARA	0.561	9.182	0.000	0.459	0.315	0.312
TR -> VC	-0.275	4.800	0.000	0.102	0.813	0.808
ARA -> VC	0.110	2.116	0.034	0.018		
DC -> VC	-0.385	4.440	0.000	0.083		
SCA -> VC	0.236	4.110	0.000	0.065		
TR -> SCA	0.008	0.562	0.574			
-> VC						
ARA -> SCA	0.063	3.172	0.002			
-> VC						
DC -> SCA	0.131	3.436	0.001			
-> VC						
TT x SCA	0.083	1.242	0.214	0.009		
-> VC						
TT x TR -> VC	0.007	0.195	0.845	0.000		
TT x ARA	0.090	1.865	0.062	0.013		
-> VC						
TT x DC -> VC	-0.140	2.236	0.025	0.019		

insights.

Sustainable competitive advantage plays a crucial mediating role, particularly in enhancing the impact of AI adoption and dynamic capabilities on value creation. The mediation effect is significant for both AI adoption ($\beta = 0.063$, $p = 0.002$) and dynamic capability ($\beta = 0.131$, $p = 0.001$), underscoring SCA's importance in translating these factors into tangible value.

Technological turbulence (TT) moderates these relationships, particularly weakening the positive impact of dynamic capabilities on value creation ($\beta = -0.140$, $p = 0.025$). This suggests that in highly turbulent technological environments, the effectiveness of dynamic capabilities may diminish.

In summary, the findings illustrate the complex interplay between technology readiness, AI adoption, dynamic capability, and sustainable competitive advantage in driving value creation. While technological readiness is crucial for AI adoption, its direct effect on value creation is nuanced, requiring strategic alignment. SCA is essential for ensuring that AI and dynamic capabilities lead to significant value creation, especially under conditions of technological turbulence. These insights

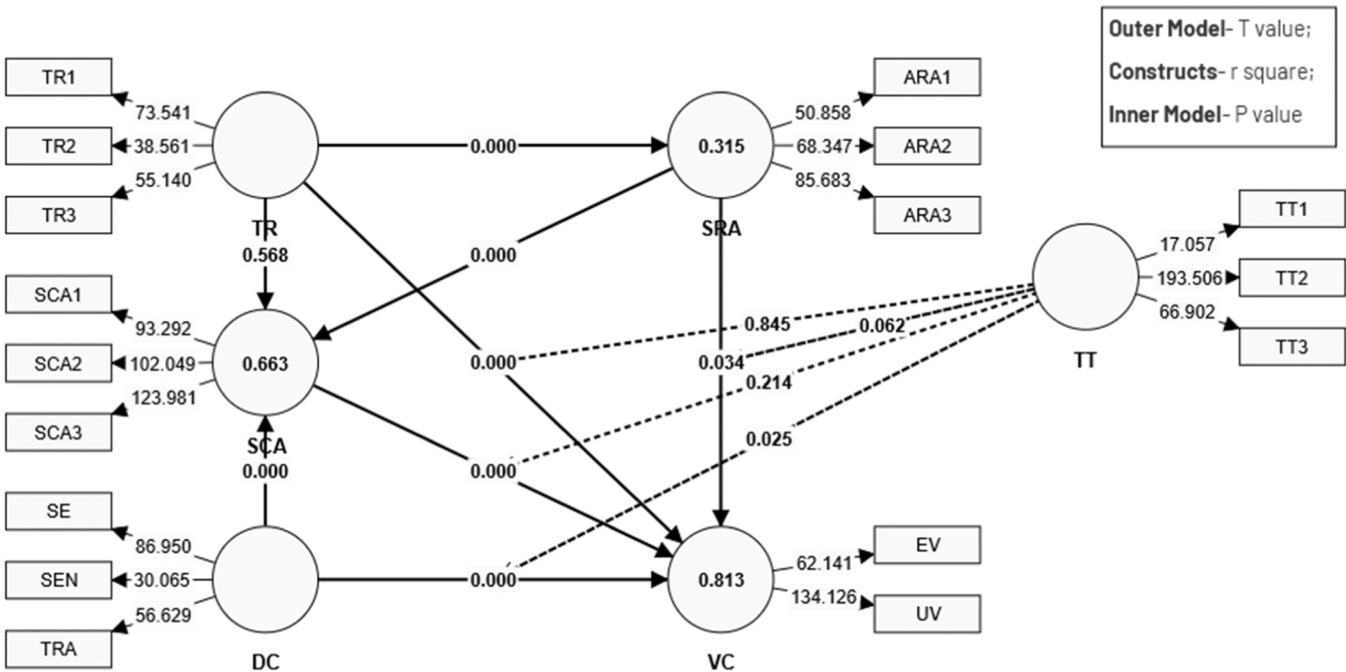


Fig. 3. Structural Model.

provide a refined understanding of how AI and strategic capabilities interact to drive value in the hospitality industry.

4.3. Discussion and hypothesis testing

The findings of this study provide a nuanced understanding of how technology readiness (TR), AI adoption (ARA), dynamic capabilities (DC), sustainable competitive advantage (SCA), and technological turbulence (TT) interact to influence value creation (VC) within the Malaysian hospitality industry. Each hypothesis tested contributes to a broader understanding of the complex dynamics at play (see Table 7).

Firstly, the strong positive influence of technology readiness on AI adoption ($\beta = 0.561, p < 0.001$) confirms the foundational role that technological preparedness plays in the successful implementation of AI. This finding aligns with prior studies by Damerji and Salimi (2021) and Godoe and Johansen (2012), which emphasize that organizations with a higher degree of technological readiness are better positioned to adopt and integrate AI technologies effectively. As highlighted by Parvez, (2020), Warner and Wäger (2019), many hospitality firms have not yet reached full AI maturity, largely because they are still developing the technology infrastructure and human capital necessary to support AI systems. This underscores that technology readiness is not just a preliminary step but an ongoing condition necessary for achieving true AI maturity. It also resonates with the motivation raised earlier—where the lack of full-scale AI maturity was identified as a central challenge—and reinforces the notion that investments in technology infrastructure and skills development are critical precursors to AI adoption.

However, the study presents a counterintuitive finding in Hypothesis 2, where technology readiness negatively impacts value creation ($\beta = -0.275, p < 0.001$). This result suggests that the process of becoming

technologically ready may initially create disruptions that negatively affect value creation (Alam et al., 2023; K. Zhu et al., 2004), likely due to the transitional challenges involved in aligning new technologies with existing business processes and customer expectations. This contrasts with the general assumption that technology readiness inherently boosts value creation, indicating that the benefits of technology readiness may be more complex and delayed than previously understood (Li et al., 2023). This finding shed light on why organizations may struggle to realize immediate value from their AI initiatives. While readiness is necessary, it does not guarantee instantaneous performance gains. Instead, it suggests that the early stages of AI deployment—marked by adjustments in infrastructure and workforce training—are crucial yet often turbulent phases that must be managed carefully to eventually achieve the value promised by AI.

AI adoption itself was found to have a modest positive effect on value creation ($\beta = 0.110, p = 0.034$), supporting the view that AI can enhance organizational efficiency and customer satisfaction when effectively implemented. This finding is consistent with literature that highlights AI's potential to streamline operations and improve decision-making, thereby adding value (Li et al., 2023; Swan et al., 2023). This outcome directly addresses the research gaps highlighted demonstrating how, once properly integrated, AI can fulfill its promise of enhancing service quality and personalizing guest experiences—objectives emphasized by scholars and industry reports (e.g., Bulchand-Gidumal et al., 2023; Guo et al., 2024).

Interestingly, dynamic capabilities, which are often viewed as essential for sustaining competitive advantage and driving value creation, showed a negative relationship with value creation ($\beta = -0.385, p < 0.001$). This suggests that while dynamic capabilities are theoretically beneficial, their implementation might initially be associated with disruptions or inefficiencies that can reduce perceived value. These findings challenge conventional wisdom and suggest that organizations may face significant "growing pains" when trying to leverage dynamic capabilities for value creation, a process that might involve substantial realignment of resources and strategies (Ahmed and Aziz, 2024; Dyduch et al., 2021; Rashidirad and Salimian, 2020b). This echoes the complexities concerning the alignment of organizational processes and strategic functions with rapidly evolving AI technologies. Dynamic capabilities are not instant panaceas; they require iterative learning and adaptation before they translate into measurable value.

Sustainable competitive advantage (SCA) was identified as a crucial mediator in the relationship between AI adoption and value creation ($\beta = 0.063, p = 0.002$), as well as between dynamic capabilities and value creation ($\beta = 0.131, p = 0.001$). These results underscore the importance of SCA as a bridge between advanced technological adoption and tangible value creation outcomes. This aligns with research by Gupta and Benson (2011) and Hariastuti et al. (2021), who argue that sustainability-focused competitive strategies enhance organizational performance and value creation. However, the expected mediation of SCA between technology readiness and value creation (H6) was not supported ($\beta = 0.008, p = 0.574$), suggesting that simply being technologically ready does not automatically lead to value creation through SCA. In line with the initial motivations, this implies that while SCA can enhance the payoff from AI adoption and dynamic capabilities, it does not automatically translate the mere existence of technological infrastructure and know-how into added value. Organizations must enact strategic and managerial interventions to harness the benefits of their readiness fully.

Regarding technological turbulence (TT), the findings were mixed. While TT significantly moderated the relationship between dynamic capabilities and value creation (H12: path coefficient = $-0.140, p = 0.025$)—highlighting the importance of adaptability in turbulent environments—it did not significantly moderate the impact of SCA, TR, or ARA on VC (H9: $p = 0.214$; H10: $p = 0.845$; H11: $p = 0.062$). These non-significant findings in H9 to H11 offer limited theoretical contribution, as they do not substantively alter our understanding of how SCA,

Table 7
Major findings of this study.

Hypothesis	Original sample	P values	Support
H1 Technology readiness positively influences AI adoption within organizations.	0.561	0.000	Yes
H2 Technology readiness positively impacts value creation for both enterprises and users.	−0.275	0.000	No
H3 AI adoption will positively influence value creation.	0.110	0.034	Yes
H4 Dynamic capabilities will positively influence value creation.	−0.385	0.000	No
H5 Sustainable competitive advantage will mediate the positive relationship between technology readiness and value creation.	0.236	0.000	Yes
H6 Sustainable Competitive Advantage mediates the influence of technology readiness on Value Creation.	0.008	0.574	No
H7 Sustainable Competitive Advantage mediates the influence of AI adoption on Value Creation.	0.063	0.002	Yes
H8 Sustainable Competitive Advantage mediates the influence of Dynamic Capability on Value Creation.	0.131	0.001	Yes
H9 Technological Turbulence moderates the relationship between Sustainable Competitive Advantage and Value Creation.	0.083	0.214	No
H10 Technological Turbulence moderates the relationship between technology readiness and Value Creation.	0.007	0.845	No
H11 Technological Turbulence moderates the relationship between AI adoption and Value Creation.	0.090	0.062	No
H12 Technological Turbulence moderates the relationship between Dynamic Capability and Value Creation.	−0.140	0.025	Yes

TR, or ARA functions under turbulent technological conditions (Ahmed et al., 2022; Senathirajah et al., 2024). Instead, they reinforce the idea that turbulence's effects are highly context-dependent and may not uniformly influence all strategic or capability-driven pathways. Still, the result for H12 supports the notion that when faced with rapid technological changes, the ability to reconfigure and reposition through dynamic capabilities becomes particularly valuable. This insight resonates with literature suggesting that firms equipped with strong adaptive capacities can better leverage turbulent conditions to innovate and create value (Lee and Yoo, 2021).

Literature on technological turbulence supports the notion that it can either hinder or enhance strategic business processes and value creation. For instance, Lee and Yoo (2021) have demonstrated how technological turbulence can positively impact internal market orientation processes and drive eco-organizational innovation and social performance. These findings align with the observed moderation in H12, suggesting that dynamic capabilities are particularly crucial in adapting and thriving in turbulent environments. Conversely, works by Cheah et al. (2018) and Matraves and Rondi (2005) indicate that industry turbulence can enhance a firm's sustainable competitive advantage and affect the sustainability of product differentiation, although these effects were not evident here.

Overall, the study's evidence clarifies the conditions under which technology readiness translates into effective AI adoption and, eventually, value. In doing so, it addresses the earlier lack of clarity about how technology preparedness and strategic capabilities interplay in delivering tangible benefits. Collectively, the findings encourage hospitality firms to invest not only in cutting-edge technologies but also in the managerial acumen, organizational structures, and long-term strategic planning necessary to navigate the path to AI maturity and achieve sustained competitive advantages.

5. Implications of this study

5.1. Theoretical implications

This study, grounded in the Resource-Based View (RBV), Dynamic Capabilities Framework, and Contingency Theory, offers significant theoretical contributions to understanding AI adoption and its impact on value creation in the hospitality industry. The findings confirm that Technology Readiness and AI Adoption are pivotal in enhancing Value Creation, aligning with the RBV's assertion that well-managed, unique resources lead to superior organizational performance (Guo et al., 2024; Bulchand-Gidumal et al., 2023). However, the study extends this theory by highlighting that the mere possession of technological assets is insufficient. Instead, the strategic integration and effective utilization of these resources are crucial for realizing their potential benefits. This nuanced understanding adds depth to the RBV by emphasizing that resources must be aligned with organizational strategies to drive value creation, a challenge also identified in earlier studies (Parvez, 2020; Warner and Wäger, 2019).

Moreover, the results underscore that organizations are yet to fully realize AI maturity largely because such maturity is contingent upon high levels of technology readiness. As introduced in Section 1, even leading industry players like Marriott and Hilton, despite their early and visible investments in AI (Al-Hyari et al., 2023; Takyar, 2024), have faced ongoing hurdles in seamlessly integrating these technologies. This finding substantiates the introductory arguments that highlight the gap between current technology readiness and the long-term, stable integration of AI-driven solutions necessary to achieve full AI maturity and sustainable competitive advantage.

Through the lens of the Dynamic Capabilities Framework, the study reveals the complexities organizations face when adapting to technological changes and market opportunities (Arias-Pérez and Cepeda-Cardona, 2022). While Dynamic Capabilities were found to moderate certain relationships, the initial negative impact on value

creation suggests that organizations encounter significant challenges in synchronizing these capabilities with strategic objectives amid technological turbulence. These difficulties echo the Introduction's discussion on how unpredictable shifts in technology landscapes demand both preparedness and agility (Calder et al., 2024; Wiggers, 2024). Thus, the present research enriches the theoretical conversation by illustrating the dual-edged nature of dynamic capabilities, resonating with the complex scenario initially outlined in the Introduction.

Finally, Contingency Theory helps elucidate the importance of fitting internal resources and capabilities to external environmental conditions. As discussed earlier (Leila, 2024; Solarova et al., 2023), the ability to navigate ethical considerations, data privacy, and customer expectations hinges on aligning organizational strategies with environmental conditions. The study confirms that organizational performance is contingent upon this strategic fit, reinforcing the Introduction's emphasis on the interplay between technology readiness, environmental turbulence, and stakeholder engagement.

5.2. Managerial insights

From a managerial perspective, the study's findings emphasize the importance of not only adopting advanced technologies such as AI but also ensuring that the organization is prepared to integrate and leverage these technologies effectively. As noted in the Introduction, many firms are at a critical juncture, attempting to move beyond basic implementation toward full-scale AI maturity (Parvez, 2020; Warner and Wäger, 2019). Managers must recognize that technology readiness—encompassing robust infrastructure, skilled personnel, and supportive organizational culture—is a foundational prerequisite for achieving this maturity and reaping the associated benefits of enhanced customer service, operational efficiency, and tailored guest experiences (Bulchand-Gidumal et al., 2023; Guo et al., 2024).

Dynamic capabilities emerge as crucial managerial levers. Managers should invest in real-time data technologies, continuous learning mechanisms, and agile decision-making processes, aligning with the Introduction's observations that market turbulence and rapid technological evolution demand organizational adaptability (Arias-Pérez and Cepeda-Cardona, 2022; Calder et al., 2024). By focusing on dynamic capabilities, leaders can navigate the transitional "growing pains" of implementing AI solutions, ultimately improving value creation as the organization's technology readiness matures.

Furthermore, as suggested in the Introduction, the concept of sustainable competitive advantage extends beyond regulatory compliance to involve strategic differentiation and building long-term stakeholder trust (Leila, 2024; Solarova et al., 2023). Managers should view sustainability as an integral component of their strategic resource configuration. By embedding ethical considerations, data privacy safeguards, and inclusivity measures (Buhalis and Moldavska, 2021a; Vuong and Tran, 2021) into AI-driven initiatives, they not only differentiate their offerings but also cultivate a strong corporate reputation over time.

In operationalizing these strategies, ongoing monitoring and evaluation through key performance indicators (KPIs) and metrics will help managers gauge their progress toward AI maturity. This iterative approach, foreshadowed by the complexity and ethical considerations raised in the Introduction, ensures that organizations can adapt their strategies in real-time, maintaining alignment with evolving market conditions and technological opportunities.

5.3. Policy implications

At the policy level, the study's findings corroborate the arguments presented in the Introduction about the urgent need for supportive policy frameworks that encourage AI adoption within the hospitality industry (Captain, 2023; Siu, 2023). Policymakers should provide incentives—such as tax breaks, grants, or favorable loans—that enhance technology readiness. By doing so, they effectively bridge the gap

between the availability of advanced technologies and their strategic, value-generating application.

The Introduction highlighted that the ethical, data privacy, and inclusivity challenges associated with AI deployment require careful regulation and capability-building (Leila, 2024; Solarova et al., 2023). Policymakers are thus encouraged to support digital literacy programs, workforce training, and educational curricula that equip employees at all levels with the necessary skills. Such measures ensure that businesses can evolve from basic AI implementation toward achieving true AI maturity, as initially emphasized in the study's motivation.

Finally, as technological turbulence intensifies, policies promoting scenario planning, risk management, and resilience strategies become indispensable. The Introduction's reference to industry disruptions and the unpredictability of technological change (Calder et al., 2024; Wiggers, 2024) resonates here. By fostering an environment where organizations can flexibly adapt to emerging challenges, policymakers help ensure that firms sustain their competitive advantages and drive long-term value creation in an ever-shifting technological landscape.

6. Conclusion and future research direction

This study has rigorously examined the complex interrelationships among technology readiness, AI adoption, dynamic capabilities, sustainable competitive advantage, and technological turbulence within the Malaysian hospitality industry. Through the theoretical lenses of the Resource-Based View, Dynamic Capabilities Framework, and Contingency Theory, this research has elucidated the pivotal roles these strategic elements play in leveraging technological assets for competitive advantage and enhanced organizational performance.

The findings from this investigation affirm that both technology readiness and AI adoption are crucial in bolstering organizational capabilities and efficiencies. Yet, the efficacy of these technological interventions transcends mere adoption, hinging significantly on their strategic integration and alignment with broader organizational strategies and prevailing market conditions. This is evidenced by the substantial roles of dynamic capabilities and the moderating influence of technological turbulence. Furthermore, the research underscores the imperative of sustainable competitive advantage in translating technological and strategic capabilities into palpable value creation, emphasizing the integration of sustainability within organizational strategy.

Looking to the future, this research opens several pathways for further inquiry. Initially, subsequent studies might explore the longitudinal impacts of AI adoption and technology readiness on value creation, particularly examining how these effects evolve as organizations mature in integrating these technologies within their operational frameworks. Additionally, there is a pertinent need to delineate the specific types of AI technologies that yield significant benefits across various domains of hospitality management, ranging from customer service enhancements to operational efficiency improvements.

Another fertile area for future research involves the exploration of cultural and organizational variables that influence the interplay between dynamic capabilities and technological turbulence. Insights into how different organizational cultures and structures assimilate and capitalize on technological changes could inform more effective strategies for technological innovation management.

Moreover, given the escalating importance of sustainability in the global business environment, further investigations could focus on how environmental and social governance influences the strategic deployment of technology in fostering competitive and sustainable advantages. Probing into the nexus between sustainability practices, technological innovation, and customer satisfaction could unveil comprehensive strategies for value creation not only in the hospitality industry but across various sectors.

In conclusion, while this study contributes significant insights into the strategic management of technology in the hospitality industry, the dynamic and rapidly evolving nature of technology and market

conditions necessitates ongoing scholarly exploration. Future research should endeavor to expand upon the findings of this study, exploring innovative dimensions and theoretical frameworks that can assist organizations in navigating the complexities of integrating technology and strategy to achieve sustainable growth and competitive edge.

Abbreviation

ARA→ Artificial Intelligence Adoption, EV→Enterprise Value, SCA→ Sustainable Competitive Advantage, SE→ Seizing, SEN→ Sensing, TR→ Transforming, TRA→ Technology Readiness, TT→ Technological Turbulence, UV→ User Value, VC→ Value Creation, DC→ Dynamic Capability

CRediT authorship contribution statement

Kokash Husam Ahmad: Visualization, Methodology, Investigation. **Alam Syed Shah:** Writing – original draft, Supervision, Conceptualization. **Ahmed Saif:** Writing – review & editing, Software, Project administration, Formal analysis. **Ahsan Mst Nilufar:** Writing – review & editing, Resources, Investigation, Data curation.

Declaration of Competing Interest

The authors declare no conflict of interest

Acknowledgement

The authors would like to thank Prince Sultan University for their support.

Funding

This research received no external funding.

Institutional review board statement

Ethical review and approval were waived for this study, due to the fact that this study was conducted as per the guidelines of the Declaration of Helsinki. The research questionnaire was anonymous, and no personal information was gathered.

Informed consent statement

Oral consent was obtained from all individuals involved in this study.

Data availability

Data will be made available on request.

References

- Abedin, B., 2022. Managing the tension between opposing effects of explainability of artificial intelligence: a contingency theory perspective. *Internet Res.* 32 (2), 425–453. <https://doi.org/10.1108/INTR-05-2020-0300>.
- Acosta-Prado, J., Tafur-Mendoza, A.A., 2022. Examining the mediating role of dynamic capabilities in the relationship between information and communication technologies and sustainable performance. *VINE J. Inf. Knowl. Manag. Syst.* 54 (3), 561–577. <https://doi.org/10.1108/VJKMS-10-2021-0257>.
- Adner, R., Zemsky, P.B., 2004. A demand based view of sustainable competitive advantage. *SSRN Electron. J.*
- Ahmed, S., Aziz, N.A., 2024. Impact of AI on customer experience in video streaming services: a focus on personalization and trust. *Int. J. Hum. –Comput. Interact.* 1–20. <https://doi.org/10.1080/10447318.2024.2400395>.
- Ahmed, S., Khalil, I., Chowdhury, B., Haque, R., Rahman, A., Senathirajah, S., Din, O., 2022. Motivators and barriers of Artificial Intelligent (AI) based teaching. *Eurasia J. Educ. Res.* 100, 74–89. <https://doi.org/10.14689/ejer.2022.100.006>.
- Alam, S.S., Ahmed, S., Kokash, H.A., 2023. Interplay of perceived organizational and external e-readiness in the adoption and integration of augmented reality and virtual

- reality technologies in Malaysian higher education institutions. *Educ. Inf. Technol.* <https://doi.org/10.1007/s10639-023-12428-7>.
- Al-Hyari, H.S., Al-Smadi, H.M., Weshah, S.R., 2023. The impact of artificial intelligence (AI) on guest satisfaction in hotel management: an empirical study of luxury hotels. *Geoj. Tour. Geosites* 48 (2spl). <https://doi.org/10.30892/gtg.482spl15-1081>.
- AlSheibani, S., Cheung, Y., & Messom, C. (2018). Artificial Intelligence Adoption: AI-readiness at Firm-Level. *Twenty-Second Pacific Asia Conference on Information Systems, Japan*.
- Alzaabi, M., & Shuhaiber, A. (2022). The role of the AI availability and perceived risks on AI adoption and organizational values. *Proceedings of the 5th International Conference on Intelligent Human Systems Integration (IHSI 2022) Integrating People and Intelligent Systems, February 22–24, 2022, Venice, Italy*.
- Arias-Pérez, J., Cepeda-Cardona, J., 2022. Knowledge management strategies and organizational improvisation: what changed after the emergence of technological turbulence caused by artificial intelligence? *Balt. J. Manag.* 17 (2), 250–265. <https://doi.org/10.1108/BJM-01-2021-0027>.
- Autry, C.W., Grawe, S.J., Daugherty, P.J., Richey, R.G., 2010. The effects of technological turbulence and breadth on supply chain technology acceptance and adoption. *J. Oper. Manag.* 28 (6), 522–536.
- Awamleh, F., Bustami, A., 2022. Examine the mediating role of the information technology capabilities on the relationship between artificial intelligence and competitive advantage during the COVID-19 pandemic. *SAGE Open* 12 (3).
- Ayegba, J.O., Lin, Z.L., 2020. A review on dynamic capacities in strategic management. *Int. Entrep. Rev.* 6 (1), 47–61.
- Balasubramanian, K., Ragavan, N.A., 2019. What are the key challenges faced by the Malaysian hospitality and tourism industry in the context of industrial revolution 4.0? *Worldw. Hosp. Tour. Themes* 11 (2), 194–203.
- Bari, N., Chimbundu, R., Chan, K.C., 2024. Interrelation between sustainable dynamic capabilities, corporate sustainability, and sustained competitive advantage. *Sustainability* 16 (7), 2864.
- Bassano, C., Barile, S., Saviano, M., Pietronudo, M.C., Cosimato, S., 2020. AI Technologies & Value Co-Creation in Luxury Context. *Proceedings of the Annual Hawaii International Conference on System Sciences*.
- Bhattacharjee, M., 2018. Disruptive value creation in business: a case based analytical study to understand the potential impact of disruptive value on competitive forces. *Manag. Today* 8 (3), 247–254.
- Buhalis, D., Harwood, T., Bogicevic, V., Viglia, G., Beldona, S., Hofacker, C., 2019. Technological disruptions in services: lessons from tourism and hospitality. *J. Serv. Manag.* 30 (4), 484–506. <https://doi.org/10.1108/JOSM-12-2018-0398>.
- Buhalis, D., Lin, M.S., Leung, D., 2022. Metaverse as a driver for customer experience and value co-creation: implications for hospitality and tourism management and marketing. *Int. J. Contemp. Hosp. Manag.* 35 (2), 701–716.
- Buhalis, D., Moldavska, I., 2021a. In-room voice-based AI digital assistants transforming on-site hotel services and guests' experiences. In *Information and Communication Technologies in Tourism 2021*. Springer International Publishing, pp. 30–44. https://doi.org/10.1007/978-3-030-65785-7_3.
- Buhalis, D., Moldavska, I., 2021b. Voice assistants in hospitality: using artificial intelligence for customer service. *J. Hosp. Tour. Technol.* 13 (3), 386–403.
- Buhalis, D., Sinarta, Y., 2019. Real-time co-creation and nowness service: lessons from tourism and hospitality. *J. Travel Tour. Mark.* 36 (5), 563–582. <https://doi.org/10.1080/10548408.2019.1592059>.
- Bulchand-Gidumal, J., William Secin, E., O'Connor, P., Buhalis, D., 2023. Artificial intelligences impact on hospitality and tourism marketing: exploring key themes and addressing challenges. *Curr. Issues Tour.* 27 (14), 2345–2362.
- Cain, L., Thomas, J.H., Alonso, M., 2019. From sci-fi to sci-fact: the state of robotics and AI in the hospitality industry. *J. Hosp. Tour. Technol.* 10 (4), 624–650. <https://doi.org/10.1108/JHTT-07-2018-0066>.
- Calder, N., Solis, F., Bell, R., 2024. March 19). Overcoming Common Challenges to Disruptive Innovation. *Harv. Bus. Rev.* <https://hbr.org/2024/03/overcoming-common-challenges-to-disruptive-innovation>.
- Campo, S., M. Díaz, A., Yagüe, M. J., 2014. Hotel innovation and performance in times of crisis. *Int. J. Contemp. Hosp. Manag.* 26 (8), 1292–1311.
- Captain, S. (2023, May 14). How AI Will Change the Workplace. *The Wall Street Journal*. <https://www.wsj.com/articles/how-ai-change-workplace-af2162ee>.
- Carvalho, P., Alves, H., 2022. Customer value co-creation in the hospitality and tourism industry: a systematic literature review. *Int. J. Contemp. Hosp. Manag.* 35 (1), 250–273.
- Celuch, K., Kasouf, C.J., 2014. An Exploration of Relationships Among Perceived Market Orientation, Strategic Flexibility, and Customer Value in the Supply Chain Under Different Conditions of Environmental Turbulence. In *Developments in Marketing Science: Proceedings of the Academy of Marketing Science*. Springer International Publishing, p. 315.
- Chandra, B., Rahman, Z., 2024. Artificial intelligence and value co-creation: a review, conceptual framework and directions for future research. *J. Serv. Theory Pract.* 34 (1), 7–32. <https://doi.org/10.1108/JSTP-03-2023-0097>.
- Chatterjee, S., Chaudhuri, R., Kamble, S., Gupta, S., Sivarajah, U., 2023. Adoption of Artificial Intelligence and Cutting-Edge Technologies for Production System Sustainability: A Moderator-Mediation Analysis. *Inf. Syst. Front.* 25 (5), 1779–1794. <https://doi.org/10.1007/s10796-022-10317-x>.
- Chatterjee, S., Rana, N.P., Tamilmani, K., Sharma, A., 2021. The effect of AI-based CRM on organization performance and competitive advantage: an empirical analysis in the B2B context. *Ind. Mark. Manag.* 97, 205–219. <https://doi.org/10.1016/j.indmarman.2021.07.013>.
- Chaudhuri, R., Chatterjee, S., Kraus, S., Vrontis, D., 2022. Assessing the AI-CRM technology capability for sustaining family businesses in times of crisis: the moderating role of strategic intent. *J. Fam. Bus. Manag.* 13 (1), 46–67. <https://doi.org/10.1108/JFBM-12-2021-0153>.
- Cheah, S., Ho, Y.-P., Li, S., 2018. Business model innovation for sustainable performance in retail and hospitality industries. *Sustainability* 10 (11), 3952.
- Chng, P.-C., Cheah, J., Amran, A., 2021. Eco-innovation practices and sustainable business performance: the moderating effect of market turbulence in the Malaysian technology industry. *J. Clean. Prod.* 283, 124556.
- Chuang, S.-H., Lin, H.-N., 2015. Co-creating e-service innovations: Theory, practice, and impact on firm performance. *Int. J. Inf. Manag.* 35 (3), 277–291. <https://doi.org/10.1016/j.jinfomgt.2015.01.002>.
- Correia, R.J., Dias, J.G., Teixeira, M.S., 2020. Dynamic capabilities and competitive advantages as mediator variables between market orientation and business performance. *J. Strategy Manag.* 14 (2), 187–206.
- Dafali, S., Al-Azad, Md.S., Mohiuddin, M., Reza, M.N.H., 2023. Strategic orientations, organizational ambidexterity, and sustainable competitive advantage: mediating role of industry 4.0 readiness in emerging markets. *J. Clean. Prod.* 401, 136765.
- Damerji, H., Salimi, A., 2021. Mediating effect of use perceptions on technology readiness and adoption of artificial intelligence in accounting. *Account. Educ.* 30 (2), 107–130.
- Dayioglu, M., Kuskü, F., 2022. Understanding the renewal of value creation activities in an environment of technological turbulences. *Pressacademia*.
- Dhanu, Prakash, V., Eugene, J., 2024. Musings on modern computing in hospitality industry. *Int. J. Indian Econ. Light* 6–12. <https://doi.org/10.36713/epri16117>.
- Doborjeh, Z., Hemmington, N., Doborjeh, M., Kasabov, N., 2021. Artificial intelligence: a systematic review of methods and applications in hospitality and tourism. *Int. J. Contemp. Hosp. Manag.* 34 (3), 1154–1176. <https://doi.org/10.1108/IJCHM-06-2021-0767>.
- Drexler, N., Beckman Lapré, V., 2019. For better or for worse: shaping the hospitality industry through robotics and artificial intelligence. *Res. Hosp. Manag.* 9 (2), 117–120.
- Dyduch, W., Chudziński, P., Cyfert, S., Zastempowski, M., 2021. Dynamic capabilities, value creation and value capture: evidence from SMEs under Covid-19 lockdown in Poland. *PLOS ONE* 16 (6), e0252423.
- Eisenhardt, K.M., Martin, J.A., 2000. Dynamic capabilities: what are they? *Strateg. Manag. J.* 21 (10–11), 1105–1121.
- Elgarhy, S.D., Abou-Shouk, M., 2022. Effects of entrepreneurial orientation, marketing, and innovation capabilities, on market performance: the mediating effect of sustainable competitive advantage. *Int. J. Contemp. Hosp. Manag.* 35 (6), 1986–2004. <https://doi.org/10.1108/IJCHM-04-2022-0508>.
- Flavián, C., Pérez-Rueda, A., Belanche, D., Casaló, L.V., 2021. Intention to use analytical artificial intelligence (AI) in services – the effect of technology readiness and awareness. *J. Serv. Manag.* 33 (2), 293–320.
- Fujiyama, R.K., Montezano, L., Alfinito, S., 2022. Are we ready for new technologies? The relation between human values and technology readiness applied to M-commerce in Brazil. *Organizaç~o Soc.* 29 (102), 537–568.
- Gaur, L., Afaq, A., Singh, G., Dwivedi, Y.K., 2021. Role of artificial intelligence and robotics to foster the touchless travel during a pandemic: a review and research agenda. *Int. J. Contemp. Hosp. Manag.* 33 (11), 4079–4098.
- Ghesh, N., Alexander, M., Davis, A., 2023. The artificial intelligence-enabled customer experience in tourism: a systematic literature review. *Tour. Rev.* <https://doi.org/10.1108/TR-04-2023-0255>.
- Giannoukou, I., 2024. Revolutionizing hospitality: strategic integration of innovation management embracing technological innovation for enhanced customer experiences. *Tech. Bus. Manag.* 7, 24–39.
- Godoe, P., Johansen, T.S., 2012. Understanding adoption of new technologies: Technology readiness and technology acceptance as an integrated concept. *J. Eur. Psychol. Stud.* 3, 38.
- Goel, P., Garg, A., Sharma, A., Rana, N.P., 2022. I won't touch money because it is dirty: examining customer's loyalty toward M-payment. *Int. J. Bank Mark.* 40 (5), 992–1016. <https://doi.org/10.1108/IJBM-06-2021-0272/FULL/XML>.
- Govindan, K., 2024. How artificial intelligence drives sustainable frugal innovation: a multitheoretical perspective. *IEEE Trans. Eng. Manag.* 71, 638–655.
- Guo, Q., Mu, L., Lou, S., 2024. Revolutionizing travel experiences: An in-depth analysis of intelligent booking systems and behavioral patterns. *Intell. Decis. Technol.* 18, 1. <https://doi.org/10.3233/IDT-230625>.
- Guptam, N.J., Benson, C.C., 2011. Sustainability and competitive advantage: an empirical study of value creation. *Compét. Forum* 9 (1).
- Haftor, D.M., Costa-Climent, R., Ribeiro-Navarrete, S., 2024. Firms' use of predictive artificial intelligence for economic value creation and appropriation. *Int. J. Inf. Manag.* 79, 102836. <https://doi.org/10.1016/j.jinfomgt.2024.102836>.
- Hahn, T., Figge, F., Barkemeyer, R., 2007. Sustainable Value creation among companies in the manufacturing sector. *Int. J. Environ. Technol. Manag.* 7 (5/6), 496.
- Hair, J., Hult, G., Ringle, C., Sarstedt, M., 2022. A primer on Partial Least Squares Structural Equation Modeling (PLS-SEM), 3rd ed.). Sage, Thousand Oaks.
- Hamdi, S., Silong, A.D., Binti Omar, Z., Mohd Rasdi, R., 2016. Impact of T-shaped skill and top management support on innovation speed; the moderating role of technology uncertainty. *Cogent Bus. Manag.* 3 (1), 1153768. <https://doi.org/10.1080/23311975.2016.1153768>.
- Hariastuti, N.L.P., Pratikto, P., Santoso, P.B., Tama, I.P., 2021. Analyzing the drivers of sustainable value creation, partnership strategies, and their impact on business competitive advantages of small & medium enterprises: a PLS-model. *Eur. J. Enterp. Technol.* 13 (110), 55–66. <https://doi.org/10.15587/1729-4061.2021.228864>.
- Haseeb, M., Hussain, H.I., Kot, S., Androniceanu, A., Jermisittiparsert, K., 2019. Role of social and technological challenges in achieving a sustainable competitive advantage and sustainable business performance. *Sustainability* 11 (14). <https://doi.org/10.3390/su11143811>.

- Huang, A., Chao, Y., Velasco, E. de la M., Bilgihan, A., Wei, W., 2021. When artificial intelligence meets the hospitality and tourism industry: an assessment framework to inform theory and management. *J. Hosp. Tour. Insights* 2514–9792. <https://doi.org/10.1108/JHTI-01-2021-0021>.
- Islam, M.A., Aldaihani, F.M.F., Saatchi, S.G., 2023. Artificial intelligence adoption among human resource professionals: does market turbulence play a role? *Glob. Bus. Organ. Excell.* 42 (6), 59–74.
- Jabeen, F., Isakovic, A.A., 2018. Examining the impact of organizational culture on trust and career satisfaction in the UAE public sector: a competing values perspective. *Empl. Relat.* 40 (6), 1036–1053. <https://doi.org/10.1108/ER-02-2017-0038>.
- Jiang, Y., Wen, J., 2020. Effects of COVID-19 on hotel marketing and management: a perspective article. *Int. J. Contemp. Hosp. Manag.* 32 (8), 2563–2573. <https://doi.org/10.1108/IJCHM-03-2020-0237>.
- Jöhnk, J., Weißert, M., Wyrtek, K., 2021. Ready or not, AI comes—an interview study of organizational AI readiness factors. *Bus. Inf. Syst. Eng.* 63 (1), 5–20. <https://doi.org/10.1007/s12599-020-00676-7>.
- Kearney, T., Vize, R., Gong, T., 2018. Digitally engaged services: a multi-level perspective on technology readiness and value co-creation behaviour in higher education: an abstract. In *Developments in Marketing Science: Proceedings of the Academy of Marketing Science*. Springer International Publishing, pp. 191–192.
- Khan, S., Munir, S., Rizvi, J., Ali Shaukat, N., 2024. The influence of artificial intelligence on business value in digital strategy: a comprehensive literature review. *Scand. J. Adv. Res. Rev.* 5 (1), 11–30.
- Kim, K.M., Park, J.S., Lee, D.H., Lee, T.W., Cho, H.H., 2011. Analysis of conjugated heat transfer, stress and failure in a gas turbine blade with circular cooling passages. *Eng. Fail. Anal.* 18 (4), 1212–1222. <https://doi.org/10.1016/j.engfailanal.2011.03.002>.
- Knudsen, E.S., aholm, L., Lien, L.B., Timmermans, B., Belik, I., Pandey, S., 2021. Stability in turbulent times? The effect of digitalization on the sustainability of competitive advantage. *J. Bus. Res.* 128, 360–369.
- Kong, H., Wang, K., Qiu, X., Cheung, C., Bu, N., 2022. 30 years of artificial intelligence (AI) research relating to the hospitality and tourism industry. *Int. J. Contemp. Hosp. Manag.* 35 (6), 2157–2177.
- Kot, M., Leszczyński, G., 2022. AI-activated value co-creation. An exploratory study of conversational agents. *Ind. Mark. Manag.* 107, 287–299. <https://doi.org/10.1016/j.indmarman.2022.10.013>.
- Lada, S., Chekima, B., Karim, Mohd R.A., Fabeil, N.F., Ayub, M.S., Amirul, S.M., Ansar, R., Bouteraa, M., Fook, L.M., Zaki, H.O., 2023. Determining factors related to Artificial Intelligence (AI) adoption among Malaysia's small and medium-sized businesses. *J. Open Innov.: Technol., Mark. Complex.* 9 (4). <https://doi.org/10.1016/j.joitmc.2023.100144>.
- Laszlo, C., Sherman, D., Whalen, J., 2006. *Creating Competitive Advantage: The Sustainable Value Model*. In *Management Models for Corporate Social Responsibility*. Springer, Berlin Heidelberg, pp. 336–343.
- Lee, W.-I., Chiu, Y., Chiang, M.-H., Chiu, C.-C., 2009. Technology readiness in the quality-value-loyalty chain. *Int. J. Electron. Bus. Manag.*
- Lee, S., Yoo, J., 2021. Determinants of a Firms Sustainable Competitive Advantages: Focused on Korean Small Enterprises. *Sustainability* 13 (1), 346.
- Leila, J. (2024, February 24). AI's transformative role in the hospitality industry. Deloitte LLP. <https://www.deloitte.com/uk/en/Industries/consumer/blogs/embracing-the-future-ai-transformative-role-in-hospitality.html>.
- Lew, S., Tan, G.W.-H., Loh, X.-M., Hew, J.-J., Ooi, K.-B., 2020. The disruptive mobile wallet in the hospitality industry: an extended mobile technology acceptance model. *Technol. Soc.* 63, 101430.
- Li, X., Zhou, Y., Liu, Y., Wang, X., Yuen, K.F., 2023. Psychological antecedents of telehealth acceptance: A technology readiness perspective. *Int. J. Disaster Risk Reduction* 91. <https://doi.org/10.1016/j.ijdrr.2023.103688>.
- Lin, C.-H., 2007. The effects of technology readiness on the formation of E-service value. *PICMET '07 - 2007 Portland International Conference on Management of Engineering & Technology*.
- Linde, L., Sjödin, D., Parida, V., Vincnet, J., 2021. Dynamic capabilities for ecosystem orchestration A capability-based framework for smart city innovation initiatives. *Technol. Forecast. Soc. Change* 166, 120614. <https://doi.org/10.1016/J.TECHFORE.2021.120614>.
- Low, M.P., Cham, T.-H., Chang, Y.-S., Lim, X.-J., 2021. Advancing on weighted PLS-SEM in examining the trust-based recommendation system in pioneering product promotion effectiveness. *Qual. Quant.* 1–30. <https://doi.org/10.1007/s11135-021-01147-1>.
- Ma, F., Khan, F., Khan, K.U., XiangYun, S., 2021. Investigating the impact of information technology, absorptive capacity, and dynamic capabilities on firm performance: an empirical Study, 215824402110613 SAGE Open 11 (4). <https://doi.org/10.1177/21582440211061388>.
- Madhala, P., Li, H., Helander, N., 2020. Organizational capabilities in data-driven value creation: a literature review. *Proc. 12th Int. Jt. Conf. Knowl. Discov., Knowl. Eng. Knowl. Manag.*
- Margarida, V., António, F., & António, R. (2018). Capabilities and innovation: the moderating effect of environmental turbulence. *European Journal of Applied Business and Management*.
- Mariam, S.M., Khawaja, K.F.K., Khan, H.G.A.K., 2023. Leveraging strategic maneuverability to augment business model innovation for competitive advantage during environmental turbulence. *NUST Bus. Rev.* 5 (1).
- Matraves, C., Rondi, L., 2005. Product Differentiation, Turbulence and Sustainability of Competitive Advantage.
- Mehralian, G., Sheikhi, S., Zatzick, C., Babapour, J., 2022. The dynamic capability view in exploring the relationship between high-performance work systems and innovation performance. *Int. J. Hum. Resour. Manag.* 34 (18), 3555–3584.
- Mhlanga, D., 2021. Artificial intelligence in the Industry 4.0, and its impact on poverty, innovation, infrastructure development, and the sustainable development goals: lessons from emerging economies? *Sustainability* 13 (11), 5788.
- Mikalef, P., Pateli, A., 2017. Information technology-enabled dynamic capabilities and their indirect effect on competitive performance: findings from PLS-SEM and fsQCA. *J. Bus. Res.* 70, 1–16.
- Mishra, A.N., Pani, A.K., 2020. Business value appropriation roadmap for artificial intelligence. *VINE J. Inf. Knowl. Manag. Syst.* 51 (3). <https://doi.org/10.1108/VJIKMS-07-2019-0107>.
- Navarro, S., Andreu, L., Cervera, A., 2014. Value co-creation among hotels and disabled customers: an exploratory study. *J. Bus. Res.* 67 (5), 813–818.
- Naz, S., Haider, S.A., Khan, S., Nisar, Q.A., Tehseen, S., 2023. Augmenting hotel performance in Malaysia through big data analytics capability and artificial intelligence capability. *J. Hosp. Tour. Insights*.
- Nurova, O., Freze, T., 2021. Competitive advantage of the sustainable digital economy. *E3S Web Conf.* 250, 6004.
- Ogbeibu, S., Emelifeonwu, J., Senadjki, A., Gaskin, J., Kaivo-oja, J., 2020. Technological turbulence and greening of team creativity, product innovation, and human resource management: implications for sustainability. *J. Clean. Prod.* 244. <https://doi.org/10.1016/j.jclepro.2019.118703>.
- Okafor, N.U., Alghorani, Y., Delaney, D.T., 2020. Improving data quality of low-cost IoT sensors in environmental monitoring networks using data fusion and machine learning approach. *ICT Express* 6 (3), 220–228. <https://doi.org/10.1016/j.icte.2020.06.004>.
- Palade, M., Carutasu, G., 2023. Organizational readiness for Artificial Intelligence adoption. *Sci. Bull. Politeh. Univ. Timișoara Trans. Eng. Manag.* 7 (1–2), 30–35. <https://doi.org/10.59168/FDMS6321>.
- Parvez, M.O., 2020. Use of machine learning technology for tourist and organizational services: high-tech innovation in the hospitality industry. *J. Tour. Futures* 7 (2), 240–244.
- Paschen, J., Paschen, U., Pala, E., Kietzmann, J.H., 2020. Artificial intelligence (AI) and value co-creation in B2B sales: activities, actors and resources. *Australas. Mark. J.* 29 (3), 243–251. <https://doi.org/10.1016/j.ausmj.2020.06.004>.
- Payne, E.H., Manser, Dahl, A.J., Peltier, J., 2021. Digital servitization value co-creation framework for AI services: a research agenda for digital transformation in financial service ecosystems. *J. Res. Interact. Mark.* 15 (2), 200–222. <https://doi.org/10.1108/JRIM-12-2020-0252>.
- Perifanis, N.-A., Kitsios, F., 2023. Investigating the influence of artificial intelligence on business value in the digital era of strategy: a literature review. *Information* 14 (2), 85.
- Pernsteiner, C., Rauseo, N., 2000. Transforming the hospitality industry into e-business. *Hosp. Rev.* 18 (2). https://digitalcommons.fiu.edu/hospitalityreview/vol18/iss2/27utm_source=digitalcommons.fiu.edu%2Fhospitalityreview%2Fvol18%2Fiss2%2F2&utm_medium=PDF&utm_campaign=PDFCoverPages.
- Pillai, R., Sivathanu, B., 2020. Adoption of AI-based chatbots for hospitality and tourism. *Int. J. Contemp. Hosp. Manag.* 32 (10), 3199–3226. <https://doi.org/10.1108/IJCHM-04-2020-0259>.
- Pisano, G.P., 1994. Knowledge, integration, and the locus of learning: an empirical analysis of process development. *Strateg. Manag. J.* 15 (1 S), 85–100. <https://doi.org/10.1002/SMJ.4250150907>.
- Pratono, A., 2021. Reinterpreting excellence for sustainable competitive advantage: the role of entrepreneurial culture under information technological turbulence. *Meas. Bus. Excell.* 26 (2), 180–196. <https://doi.org/10.1108/MBE-04-2021-0056>.
- Rana, N.P., Chatterjee, S., Dwivedi, Y.K., Akter, S., 2022. Understanding dark side of artificial intelligence (AI) integrated business analytics: assessing firm's operational inefficiency and competitiveness. *Eur. J. Inf. Syst.* 31 (3), 364–387. <https://doi.org/10.1080/0960085X.2021.1955628>.
- Rashidirad, M., Salimian, H., 2020a. SMEs dynamic capabilities and value creation: the mediating role of competitive strategy. *Eur. Bus. Rev.* 32 (4), 591–613.
- Rashidirad, M., Salimian, H., 2020b. SMEs dynamic capabilities and value creation: the mediating role of competitive strategy. *Eur. Bus. Rev.* 32 (4), 591–613.
- Rodriguez, M.A., Ricart, J.E., Sanchez, P., 2002. Sustainable development and the sustainability of competitive advantage: a dynamic and sustainable view of the firm. *Creat. Innov. Manag.* 11 (3), 135–146.
- Roy, B.K., Pagalditvi, S.R., 2023. Advancements in arena technology: enhancing customer experience and employee adaptation in the tourism and hospitality industry. *Smart Tour.* 4 (1), 2330.
- Sasmoko, S., Mihardjo, L.W.W., Alamsjah, F., Elidjen, E., 2019. Dynamic capability: the effect of digital leadership on fostering innovation capability based on market orientation. *Manag. Sci. Lett.* 1633–1644.
- Senathirajah, A.R., bin, S., Alainati, S., Haque, R., Ahmed, S., Khalil, Md.I., Chowdhury, B., 2024. Antecedents and consequence of trust-commitment towards artificial based customer experience. *UCJC Bus. Soc. Rev.* 696–777. <https://doi.org/10.3232/UBR.2024.V21.N1.16>.
- Senbeto, D., Hon, A.H.Y., 2021. Development of employees resilience in technologically turbulent environments: probing the mechanisms of consonance–dissonance and crisis leadership. *Int. J. Contemp. Hosp. Manag.* 33 (8), 2721–2740. <https://doi.org/10.1108/IJCHM-11-2020-1263>.
- Shehadeh, E., Rawashdeh, A., Rababah, A., AlOkdeh, S.K., 2022. Adoption of robotic process Automation (RPA) and its effect on business value: an internal auditors perspective. *J. Posit. Sch. Psychol.* 6 (6), 9832–9847.
- Shiau, W.-L., Sarstedt, M., Hair, J.F., 2019. Internet research using partial least squares structural equation modeling (PLS-SEM). *Internet Res.* 29 (3), 398–406. <https://doi.org/10.1108/IntR-10-2018-0447>.

- Siu, E., 2023. AI may not replace you, but someone who uses it can — here's the No. 1 skill you need to stay relevant. CNBC. <https://www.cnbc.com/2024/05/13/ai-skill-you-must-learn-to-stay-competitive-in-the-jobs-market.html>.
- Solakis, K., Katsoni, V., Mahmoud, A.B., Grigoriou, N., 2022. Factors affecting value co-creation through artificial intelligence in tourism: a general literature review. *J. Tour. Futures* 10 (1), 116–130.
- Solarova, S., Podroužek, J., Mesarčík, M., Gavornik, A., Bielikova, M., 2023. Reconsidering the regulation of facial recognition in public spaces. *AI Ethics* 3 (2). <https://doi.org/10.1007/s43681-022-00194-0>.
- Sudaryanto, M.R., Hendrawan, M.A., Andrian, T., 2023. The effect of technology readiness, digital competence, perceived usefulness, and ease of use on accounting students artificial intelligence technology adoption. *E3S Web Conf.* 388, 4055.
- Sun, Y., Xu, X., Yu, H., Wang, H., 2022. Impact of value cocreation in the artificial intelligence innovation ecosystem on competitive advantage and innovation intelligibility. *Syst. Res. Behav. Sci.* 39 (3), 474–488.
- Sun, X., Zhang, Q., 2021. How can dynamic capabilities make sense in avoiding value co-creation traps? *Manag. Decis.* 60 (3), 735–757.
- Swan, E.L., Peltier, J.W., Dahl, A.J., 2023. Artificial intelligence in healthcare: the value co-creation process and influence of other digital health transformations. *J. Res. Interact. Mark.* 18 (1), 109–126.
- Takyar, A., 2024. AI in hospitality: use cases, applications, solution and implementation. LeewayHertz - AI Development Company. (<https://www.leewayhertz.com/ai-use-cases-in-hospitality/>).
- Talebnejad, A., 2012. The role and effect of information technology in the creation and maintenance of sustainable competitive advantage. *Int. J. Inf. Sci. Technol.* 6 (1).
- Thai, Q.H., Mai, K.N., 2023. Does entrepreneurial financial support guarantee new ventures' performance via competitive advantage and innovation? Empirical answers from Ho Chi Minh City Region, Vietnam. *Sustain. (Switz.)* 15 (21). <https://doi.org/10.3390/su152115519>.
- Thompson, C.G., Kim, R.S., Aloe, A.M., Becker, B.J., 2017. Extracting the variance inflation factor and other multicollinearity diagnostics from typical regression results. *Basic Appl. Soc. Psychol.* 39 (2), 81–90. <https://doi.org/10.1080/01973533.2016.1277529>.
- Tuyet, T.T., Tuan, N.M., 2020. The relationships of technology readiness, perceived value, satisfaction, and continuance intention – a study of self-service technologies in Viet Nam. *Science & Technology Development. J. - Econ. - Law Manag.* 3, SI24–SI34.
- Vuong, H.Q., & Tran, T.M. (2021, October 21). The relationship between innovation and value creation by artificial intelligence: The case of the tourism industry in the COVID-19 pandemic. 2nd International Conference on Human-Centered Artificial Intelligence (Computing4Human 2021).
- Wang, M.L., 2023. Effects of the green finance policy on the green innovation efficiency of the manufacturing industry: a difference-in-difference model. *Technol. Forecast. Soc. Change* 189. <https://doi.org/10.1016/J.TECHFORE.2023.122333>.
- Warner, K.S.R., Wäger, M., 2019. Building dynamic capabilities for digital transformation: an ongoing process of strategic renewal. *Long. Range Plan.* 52 (3), 326–349. <https://doi.org/10.1016/j.lrp.2018.12.001>.
- Washizaki, H., 2020. Towards Software Value Co-Creation with AI. In W.K. Chan, B. Claycomb, H. Takakura, J.J. Yang, Y. Teranishi, D. Towey, S. Segura, H. Shahriar, S. Reisman, & S.I. Ahamed (Eds), 2020 IEEE 44th Annual Computers, Software, And Applications Conference (COMPSAC 2020) (pp. 1117–1118). <https://doi.org/10.1109/COMPSAC48688.2020.0-112>.
- Whang, S.E., Roh, Y., Song, H., Lee, J.-G., 2023. Data collection and quality challenges in deep learning: a data-centric AI perspective. *Vldb J.* 32 (4), 791–813. <https://doi.org/10.1007/s00778-022-00775-9>.
- White, C.J., McMurray, A.J., Rudito, P., 2017. Human values and technology readiness: the mediating role of consumer perceived value. *Int. J. Serv. Technol. Manag.* 23 (4), 331.
- Wiesel, T., 2022. Value Creation Measurement and Management in Times of Radical Social and Technological Change. *J. Creat. Value* 8 (2), 171–183.
- Wiggers, K., 2024. This Week in AI: AI isn't world-ending — but it's still plenty harmful. Tech Crunch. <https://techcrunch.com/2024/08/14/this-week-in-ai-ai-isnt-world-ending-but-its-still-plenty-harmful/>.
- Xu, W. (2023, August 16). Analysis of AI Virtual Human Industry Value Creation in the Context of the Internet. Proceedings of the 2023 2nd International Conference on Urban Planning and Regional Economy (UPRE 2023). https://doi.org/https://doi.org/10.2991/978-94-6463-218-7_23.
- Yieh, K., Chen, J., Wei, M.B., 2012. The effects of technology readiness on customer perceived value: an empirical analysis. *J. Fam. Econ. Issues* 33 (2), 177–183.
- Yu, Z., 2021. The effect of teacher presence in videos on intrinsic cognitive loads and academic achievements. *Innov. Edu. Teach. Int.* 59 (5), 574–585.
- Zahra, S.A., 2021. The resource-based view, resourcefulness, and resource management in startup firms: a proposed research agenda. *J. Manag.* 47 (7), 1841–1860. <https://doi.org/10.1177/01492063211018505>.
- Zhang, X., Chu, Z., Ren, L., Xing, J., 2023. Open innovation and sustainable competitive advantage: the role of organizational learning. *Technol. Forecast. Soc. Change* 186, 122114.
- Zhu, K., Kraemer, K.L., Xu, S., Dedrick, J., 2004. Information technology payoff in E-Business environments: an international perspective on value creation of E-Business in the financial services industry. *J. Manag. Inf. Syst.* 21 (1), 17–54. <https://doi.org/10.1080/07421222.2004.11045797>.
- Zhu, Y., Wang, P., Duan, W., 2022. Exploration on the core elements of value co-creation driven by ai-measurement of consumer cognitive attitude based on Q-methodology. *Front. Psychol.* 13. <https://doi.org/10.3389/fpsyg.2022.791167>.